

Decent Coaching Classes

VIII CBSE

Tuesday Holiday Homework

Understanding Quadrilaterals — Parallelogram & Rhombus

Section A — Multiple Choice Questions (Tick the correct option)

1. In a parallelogram, opposite angles are:
(a) Equal (b) Supplementary (c) Complementary (d) Unequal
2. The diagonals of a rhombus:
(a) Are equal (b) Bisect each other at right angles (c) Do not bisect each other (d) Are parallel
3. If one angle of a parallelogram is 70° , its adjacent angle is:
(a) 70° (b) 110° (c) 90° (d) 180°
4. A rhombus is a parallelogram in which:
(a) All angles are equal (b) All sides are equal (c) Diagonals are equal (d) One angle is 90°
5. The sum of all interior angles of a quadrilateral is:
(a) 180° (b) 270° (c) 360° (d) 450°

Section B — Fill in the Blanks

6. In a parallelogram, opposite sides are _____.
7. The diagonals of a parallelogram _____ each other.
8. A rhombus has all four sides _____ in length.
9. Each diagonal of a rhombus divides it into two _____ triangles.
10. The sum of any two adjacent angles of a parallelogram is _____.

Section C — Short Answer Questions

11. State whether the following statements are True or False:
(a) Every rhombus is a parallelogram.
(b) Every parallelogram is a rhombus.
(c) The diagonals of a rhombus are always equal in length.
(d) Opposite angles of a parallelogram are equal.
12. Two adjacent angles of a parallelogram are in the ratio 4 : 5. Find the measure of each angle.
13. In parallelogram PQRS, $\angle P = 3x - 5^\circ$ and $\angle R = 2x + 15^\circ$. Find the value of x and the measure of $\angle P$.
14. The diagonals of a rhombus are 16 cm and 12 cm. Find the length of each side of the rhombus.
15. In a parallelogram ABCD, $AB = 3y - 1$ and $CD = 2y + 4$. Find the length of AB.

Section D — Long Answer / Application Based Questions

16. The diagonals of a rhombus measure 24 cm and 18 cm. Find the area of the rhombus.
17. In a parallelogram ABCD, the diagonals AC and BD intersect at O. If $AO = 5$ cm and $BO = 7$ cm, find the lengths of AC and BD.
18. ABCD is a rhombus in which the diagonal $AC = 16$ cm and side $AB = 10$ cm. Find the length of diagonal BD.
19. Show, with the help of a suitable property, that in a parallelogram, opposite angles are equal. (Write the property used; a formal proof is not required.)
20. One angle of a parallelogram is 30° more than twice its adjacent angle. Find all four angles of the parallelogram.

ANSWER KEY

Section A — Multiple Choice Questions

1. (a) Equal
2. (b) Bisect each other at right angles
3. (b) 110° [Adjacent angles of a parallelogram are supplementary: $180^\circ - 70^\circ = 110^\circ$]
4. (b) All sides are equal
5. (c) 360°

Section B — Fill in the Blanks

6. Equal and parallel
7. Bisect
8. Equal
9. Congruent
10. 180° (supplementary)

Section C — Short Answer Questions

11. (a) True (b) False (c) False (d) True
12. Let the angles be $4x$ and $5x$. Since adjacent angles of a parallelogram are supplementary: $4x + 5x = 180^\circ \Rightarrow 9x = 180^\circ \Rightarrow x = 20^\circ$. So the angles are 80° and 100° (each pair of opposite angles repeats: $80^\circ, 100^\circ, 80^\circ, 100^\circ$).
13. Opposite angles of a parallelogram are equal, so $\angle P = \angle R$: $3x - 5 = 2x + 15 \Rightarrow x = 20$. Thus $\angle P = 3(20) - 5 = 55^\circ$.
14. Diagonals of a rhombus bisect each other at right angles, so half-diagonals are 8 cm and 6 cm. Side = $\sqrt{(8^2 + 6^2)} = \sqrt{(64+36)} = \sqrt{100} = 10$ cm.
15. Opposite sides of a parallelogram are equal: $3y - 1 = 2y + 4 \Rightarrow y = 5$. $AB = 3(5) - 1 = 14$ units.

Section D — Long Answer / Application Based Questions

16. Area of rhombus = $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 24 \times 18 = 216 \text{ cm}^2$.
17. Diagonals of a parallelogram bisect each other, so O is the midpoint of both diagonals: $AC = 2 \times AO = 2 \times 5 = 10$ cm; $BD = 2 \times BO = 2 \times 7 = 14$ cm.
18. Diagonals of a rhombus bisect each other at right angles. Half of $AC = 8$ cm. In right triangle AOB: $AB^2 = AO^2 + BO^2 \Rightarrow 10^2 = 8^2 + BO^2 \Rightarrow BO^2 = 100 - 64 = 36 \Rightarrow BO = 6$ cm. So $BD = 2 \times 6 = 12$ cm.
19. Property used: A diagonal of a parallelogram divides it into two congruent triangles (by SSS/ASA), so the angles at the vertices they don't share are equal — this shows opposite angles of a parallelogram are equal.
20. Let the smaller angle = x , so the adjacent angle = $2x + 30$. Adjacent angles are supplementary: $x + 2x + 30 = 180 \Rightarrow 3x = 150 \Rightarrow x = 50^\circ$. So angles are $50^\circ, 130^\circ, 50^\circ, 130^\circ$.